Dynamics and relativity

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Lecture 1 – 22/01/26

§1 The structure of the Newtonian universe

- Three dimensional space endowed with a *Cartesian reference frame* – an origin and some axes.
- Time can be labelled in an arbitrary reference frame, by a real number t.
- A *point particle* is an idealised object that is completely determined by its position at a given point in time, $\mathbf{x}(t)$. Its velocity

$$\mathbf{v} = \frac{d\mathbf{x}}{dt} = \left(\frac{dx_1}{dt}, \frac{dx_2}{dt}, \frac{dx_3}{dt}\right) = \dot{\mathbf{x}}$$

  is tangent to the trajectory. The acceleration

$$\mathbf{a} = \frac{d^2\mathbf{x}}{dt^2} = \left(\frac{d^2x_1}{dt^2}, \frac{d^2x_2}{dt^2}, \frac{d^2x_3}{dt^2}\right) = \ddot{\mathbf{x}} = \dot{\mathbf{v}}$$

The above is *not* sufficient to write down Newton's equations. Consider, for example, a free particle not experiencing any forces. The position of this particle is $\mathbf{x}(t)$, but which frame should we use?

§1.1 Law of inertia

Proposition 1.1 (Law of inertia)

There exist *inertial frames* in which a free particle has a constant velocity.

This is an improved version of Newton's first law. Indeed, it is a true statement, but not an obvious one.

§1.2 Galilean relativity principle

Definition 1.2. A *Galilean transformation* has the form

$$\mathbf{x}' = R\mathbf{x} + \mathbf{k} + \mathbf{w}t$$

where $R \in O(3)$, $\mathbf{k}$ is a translation vector, and $\mathbf{w}$ is a *boost*. In particular, $\ddot{\mathbf{x}'} = \ddot{\mathbf{x}}$.

Proposition 1.3 (Galilean relativity principle)

A frame related to an inertial frame by a *Galilean transformation* is also an inertial frame, and all laws of physics are the same in both frames.

Example of a boost: dropping a ball on a boat moving at constant velocity.

Galilean invariance also restricts the forces that are possible (see example sheet). For example, there are no 'special points' in the universe (or special directions, times, velocities, etc) – they are all relative. This is *not* the case for acceleration – an object accelerating in one inertial frame is accelerating with the same magnitude in all inertial frames.

Galilean transformations form the *Galilean group*, often supplemented by shifts in time, under which the laws of physics are also invariant. In particular, this implies that all inertial frames have the same time, called absolute time.

§2 Forces

Once there is more than one particle in the universe, there will be interactions between them. In Newtonian physics, these are described by forces.

§2.1 Newton's second law

In an inertial frame,

$$\dot{\mathbf{p}} = \mathbf{F}$$

where the *momentum* $\mathbf{p}$ is $m\mathbf{v}$, where m is the *inertial mass*, mass for short. The mass is an additional property for point particles that we take to be constant in time unless otherwise stated.

The force $\mathbf{F}$ depends on the ongoing interactions, but it can only depend (axiomatically) on $\mathbf{x}$ and $\mathbf{v}$. This implies that Newton's second law is a second-order differential equation, and that, given $\mathbf{x}$ and $\mathbf{v}$ at $t = 0$ for all particles, Newton's equations will uniquely determine $\mathbf{x}(t)$ for all future times.

We should note that Newtonian mechanics has been superseded by quantum mechanics (for small things) and general relativity (for fast things).

§2.2 Conservative forces

Definition 2.1. A force $\mathbf{F}$ is called *conservative* if it can be written

$$\mathbf{F} = -\nabla V$$

for some *potential* scalar field V.

For example, *gravitational potential energy* of a particle of mass m at $\mathbf{x}$ due to a particle of mass M at $\mathbf{x}_0$ is

$$V = -\frac{GMm}{|\mathbf{x} - \mathbf{x}_0|}$$

where G is a constant.

Now we can get the force by taking the gradient. First,

$$\partial_i \left(|\mathbf{x} - \mathbf{x}_0|^2\right) = 2|\mathbf{x} - \mathbf{x}_0|\partial_i|\mathbf{x} - \mathbf{x}_0|$$

But also,

$$\begin{aligned}
\partial_i \left(|\mathbf{x} - \mathbf{x}_0|^2\right) &= \partial_i \left((\mathbf{x} - \mathbf{x}_0)_j(\mathbf{x} - \mathbf{x}_0)_j\right) \\
&= 2(\mathbf{x} - \mathbf{x}_0)_j \partial_i(x_j - (x_0)_j) \\
&= 2(\mathbf{x} - \mathbf{x}_0)_j \delta_{ij} \\
&= 2(\mathbf{x} - \mathbf{x}_0)_i
\end{aligned}$$

and so we have

$$\begin{aligned}
\partial_i|\mathbf{x} - \mathbf{x}_0| &= \frac{(\mathbf{x} - \mathbf{x}_0)_i}{|\mathbf{x} - \mathbf{x}_0|} \\
\implies \boxed{\nabla|\mathbf{x} - \mathbf{x}_0| = \frac{\mathbf{x} - \mathbf{x}_0}{|\mathbf{x} - \mathbf{x}_0|}}
\end{aligned}$$

Hence

$$\mathbf{F} = -\nabla V = -GMm\frac{\mathbf{x} - \mathbf{x}_0}{|\mathbf{x} - \mathbf{x}_0|^3}$$

Let $\mathbf{r} = \mathbf{x} - \mathbf{x}_0$, then

$$\mathbf{F} = -\frac{GMm}{r^2}\hat{\mathbf{r}}$$

where $\hat{\mathbf{r}} = \mathbf{r}/r$ is a unit vector. Thus we have the inverse square law of gravity.

> **Remark.** Sometimes we write $V = m\Phi$ where $\Phi = -\frac{GM}{|\mathbf{x} - \mathbf{x}_0|}$ is called the *gravitational potential*.

Near the surface of the Earth, take $\mathbf{x}_0 = \mathbf{0}$, and $|\mathbf{x}| = R + z$, where R is the radius of the Earth and $z \ll R$. Then

$$\begin{aligned}
\Phi(R + z) &= -\frac{GM}{R + z} \\
&= -\frac{GM}{R}\left[1 - \frac{z}{R} + \frac{z^2}{R^2} + \cdots\right]
\end{aligned}$$

$$\approx \text{constant} + \underbrace{\frac{GM}{R^2}}_{g} z$$

where $g \approx 9.8\text{ms}^{-2}$. Since the constant will drop out when taking the gradient, we ignore it, so we have

$$\begin{aligned} &\Phi \approx gz \\ \implies &\mathbf{F} = -mg\hat{\mathbf{z}} \end{aligned}$$

This force leads to the simplest non-trivial example of motion due to a force. By Newton's second law, we have

$$m\ddot{\mathbf{x}} = m\mathbf{g}$$

where $\mathbf{g} = (0, 0, -g)$. We need only consider the z component, which gives

$$\begin{aligned} &m\ddot{z} = -mg \\ \implies &\dot{z} = v_0 - gt \\ \implies &z = z_0 + v_0 t - \frac{1}{2}gt^2 \end{aligned}$$

for an initial height z_0 and an initial velocity v_0.

§2.3 Energy

Proposition 2.2

Conservative forces have a *conserved energy*

$$\begin{aligned} E &= \frac{1}{2}m\dot{\mathbf{x}} \cdot \dot{\mathbf{x}} + V(\mathbf{x}) \\ &= \frac{1}{2}m\dot{x}_i\dot{x}_i + V(\mathbf{x}) \end{aligned}$$

Proof. We have

$$\begin{aligned} \frac{dE}{dt} &= m\dot{x}_i\ddot{x}_i + \frac{\partial V}{\partial x_i}\dot{x}_i \\ &= m\dot{x}_i\left(m\ddot{x}_i + \frac{\partial V}{\partial x_i}\right) \\ &= 0 && (m\ddot{\mathbf{x}} = -\nabla V) \end{aligned}$$

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Example 2.3

Suppose we throw an object into space and want it to never fall back down. The minimal velocity required is called the escape velocity.

When the object is thrown, its energy is

$$E = \frac{1}{2}mv^2 - \frac{GMm}{R}$$

To not fall back, the object must reach $|\mathbf{x}| \to \infty$ without the velocity going to zero. We have

$$E_\infty = \frac{1}{2}mv_\infty^2 - 0$$

By conservation of energy, $E = E_\infty$, and so to make $v_\infty > 0$ we must have

$$\begin{aligned} \frac{1}{2}mv^2 &> \frac{GMm}{R} \\ \implies v^2 &> \frac{2GM}{R} \\ \implies v &> \sqrt{\frac{2GM}{R}} \end{aligned}$$

which is approximately 10kms^{-1} for Earth.

Note: the mass cancels in v_{escape} because the *gravitational mass* (from the inverse square law) is equal to the *inertial mass* (that appears in Newton's second).

Proposition 2.4 (Principle of equivalence)

Objects have the same gravitational and inertial masses.

Remark. This is not known and there is no reason why we would expect this to be true – whether there may be tiny deviations between the masses is under investigation.

It's useful to write energy as $E = T + V$, where T is the *kinetic energy* $\frac{1}{2}m\dot{\mathbf{x}} \cdot \dot{\mathbf{x}}$.

Definition 2.5. The *work done* by a force as a particle moves along a trajectory C is given by

$$W = \int_C \mathbf{F} \cdot d\mathbf{x}$$

Proposition 2.6

For a conservative force, the work done along a trajectory only depends on the endpoints of the trajectory, not on the path itself.

Proof 1. We have

$$W = \int_C \mathbf{F} \cdot d\mathbf{x} = \int_{t_1}^{t_2} \mathbf{F} \cdot \frac{d\mathbf{x}}{dt}\, dt$$

$$\begin{aligned}
&= m \int_{t_1}^{t_2} \ddot{\mathbf{x}} \cdot \dot{\mathbf{x}}\, dt \\
&= \frac{1}{2} m \int_{t_1}^{t_2} \frac{d}{dt} (\dot{\mathbf{x}} \cdot \dot{\mathbf{x}})\, dt \\
&= T(t_2) - T(t_1) \\
&= V(t_1) - V(t_2) && (T_2 + V_2 = T_1 + V_1) \\
&= V(\mathbf{x}(t_1)) - V(\mathbf{x}(t_2)) \\
&= V(\mathbf{x}_1) - V(\mathbf{x}_2)
\end{aligned}$$

from which it's obvious that the work done only depends on the endpoints. ■

Proof 2. We have

$$\begin{aligned}
W = \int_C \mathbf{F} \cdot d\mathbf{x} &= -\int_C \nabla V \cdot d\mathbf{x} \\
&= V(x_1) - V(x_2)
\end{aligned}$$

■

§2.4 Electromagnetic forces

Forces that depend on velocity typically don't have a conserved energy (e.g. friction), but the Lorentz force is an exception.

Electromagnetic fields $\mathbf{E}$ and $\mathbf{B}$ exert the following force on a particle with *charge* q:

$$\mathbf{F} = q\left[\mathbf{E}(\mathbf{x}) + \dot{\mathbf{x}} \times \mathbf{B}(\mathbf{x})\right]$$

We will restrict to static electromagnetic fields, in which case we have

$$\mathbf{E} = -\nabla \Phi$$

for some electric potential Φ. We claim the conserved energy is $E = \frac{1}{2} m \dot{\mathbf{x}}^2 + q\Phi(\mathbf{x})$. We have

$$\begin{aligned}
\frac{dE}{dt} &= m\ddot{\mathbf{x}} \cdot \dot{\mathbf{x}} + q\nabla\Phi \cdot \dot{\mathbf{x}} \\
&= (\mathbf{F} + q\nabla\Phi) \cdot \dot{\mathbf{x}} \\
&= q(\dot{\mathbf{x}} \times \mathbf{B}) \cdot \dot{\mathbf{x}} \\
&= 0
\end{aligned}$$

The velocity-dependent force is orthogonal to the trajectory of the particle, so it does no work on the particle and energy is still conserved.

Electric forces are similar to gravitational ones: the potential Φ at $\mathbf{x}$ due to another particle of charge Q at $\mathbf{x}_0$ is

$$\Phi = \frac{Q}{4\pi\varepsilon_0} \cdot \frac{1}{|\mathbf{x} - \mathbf{x}_0|}$$

where ε_0 is the *permittivity of free space*. Just like with gravity, this leads to an inverse square law called the *Coulomb force*.

An important distinction from gravity is that charges can be positive or negative, but mass is always positive.

Magnetic forces are different. A charged particle in a magnetic field obeys

$$m\ddot{\mathbf{x}} = q\dot{\mathbf{x}} \times \mathbf{B}$$

This is a vector differential equation, which we can solve by writing out in Cartesian coordinates (we will also take $\mathbf{B}$ constant and in the z-direction, $\mathbf{B} = (0, 0, B) = B\hat{\mathbf{z}}$). The equations become

$$m\ddot{x} = q\dot{y}B \tag{1}$$
$$m\ddot{y} = -q\dot{x}B \tag{2}$$
$$m\ddot{z} = 0 \tag{3}$$

The first thing we can observe is that $z = z_0 + v_z t$, that is, the particle moves with constant velocity in the z-direction. Now, differentiating (1) and subbing into (2), we have

$$m\dddot{x} = q\ddot{y}B = -\frac{q^2\dot{x}B^2}{m}$$

and this is a second-order equation for $\dot{x}$ – we have

$$\dot{x} = \tilde{A}\sin(\omega t + \phi)$$

where $\omega = \frac{qB}{m}$ is the *cyclotron frequency*. Hence

$$\boxed{x = x_0 + A\cos(\omega t + \phi)}$$

Plugging into (1), we have

$$qB\dot{y} = -mA\omega^2\cos(\omega t + \phi)$$
$$\implies \boxed{y = y_0 - A\sin(\omega t + \phi)}$$

Hence, the particle is moving in a circle in the x-y plane while moving with a constant velocity in the z-direction.

DIAGRAM.

Alternatively, let $\xi = x + iy$, and write $(1) + i(2)$. We have

$$m\ddot{\xi} = -iqB\dot{\xi}$$

which we can solve nice and easily: $\xi = c_1 e^{-i\omega t} + c_2$. Let's put

$$c_1 = Ae^{-i\phi}$$
$$c_2 = x_0 + iy_0$$

Plugging in and taking real and imaginary parts, we recover the exact same solution as before.

> **Remark.** There is something 'complex' underlying cyclotron motion – c.f. quantum Hall effect.

§2.5 Motion in one dimension

Often, problems involving higher dimensions can be reduced to one-dimensional motion. In such a case, we have

$$m\ddot{x} = F_x$$

If F_x is independent of velocity, it can always be written as the gradient of a potential by setting

$$V(x) = -\int_{x_0}^{x} F_x(x')dx'$$

for an arbitrary reference point x_0. So the energy

$$E = \frac{1}{2}m\dot{x}^2 + V(x)$$

is always conserved. Then, putting $E =$ constant, we get a first-order differential equation for $x(t)$ which is easily integrated: we have

$$\begin{aligned}
\dot{x} &= \pm\sqrt{\frac{2}{m}(E - V(x))} \\
\implies t - t_0 &= \pm\int_{x_0}^{x} \frac{1}{\sqrt{\frac{2}{m}(E - V(x'))}}dx'
\end{aligned}$$

and so solving Newton's equations in one dimension essentially just amounts to doing this integral.